



THE UNIVERSITY OF HONG KONG 香港大學
faculty of architecture 建築學院



Logistics and Supply Chain MultiTech R&D Centre
物流及供應鏈多元技術研發中心



THE HONG KONG
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The Hong Kong Polytechnic University
香港理工大學地理空間智能研究中心

建設資產的三維點雲自動建模系統

Automated Scan-to-BIM for Precise 3D Modeling of Built Assets

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Dept of Real Estate & Construction, University of Hong Kong
National Center of Tech. Inno. for Digital Construction-HK Branch, HKU

14 Jul 2025



Who we are



◆ Faculty of Architecture

- QS World's 13th in the subject

◆ LSCM Center, Lead

◆ Three investigators

- Prof Frank Xue, *Associate Professor*. AI, BIM, Optimization, Digital Twin
- Prof Anthony Yeh, *Chair Professor*. GIS, Urban planning
- Prof Wilson Lu, *Chair Professor*. Digital construction

◆ NCTI-DC Hong Kong

- Est. Oct. 2024
- Hong Kong's first NTCI branch



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1 Smart construction & digitalization



Smart construction as a national strategy

- 中央政府：“發展智能建造”——《十四五規劃和2035年遠景目標綱要》
- 國家自然資源部：《关于全面推进实景三维中国建设的通知》
- 国务院：《数字中国建设》

In Hong Kong

- Election Manifesto of Chief Executive Election 2022
- Development Bureau's Technical Circular (2021)
- CIC's Construction Digitalization Roadmap (2021)

6 Applications (DAs)



中华人民共和国中央人民政府

Q EN

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首页 > 新闻 > 滚动

中华人民共和国国民经济和社会发展第十四个五年规划和2035年远景目标纲要



中华人民共和国自然资源部
Ministry of Natural Resources of the People's Republic of China



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發展局
工務科

香港馬利士道二號
政府總部西翼十八樓

Ref : DEVB(W) 430/00/01

Group : 2, 5, 6

20 December 2021

2022年02月24日

2022年02月24日

2022年02月24日

2022年02月24日

2022年02月24日

2022年02月24日

2022年02月24日

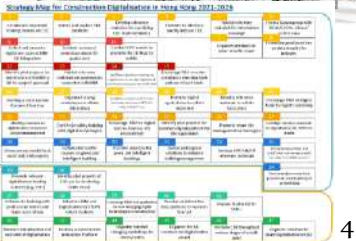
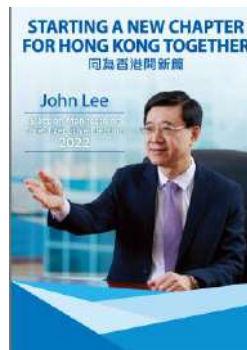
2022年02月24日

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2022年02月24日





1 BIM is a key to construction digitalization



- ◆ Local construction market: ~HK\$300* billion.
- ◆ BIM (building information modeling) in Hong Kong

- “Wider” by industry-wide mandatory uses

All public works >HK\$30M since 2020 (DevB, 2019)

All private projects >HK\$300M by 2026 (CIC, 2022)

- “Deeper” for values

“A roadmap on BIM for plans submission” (Policy Addr 2022)

20 mandatory BIM stages since 2022 (DevB, 2021)

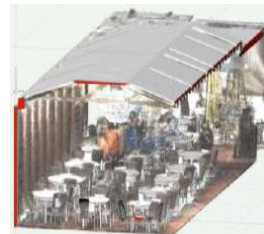
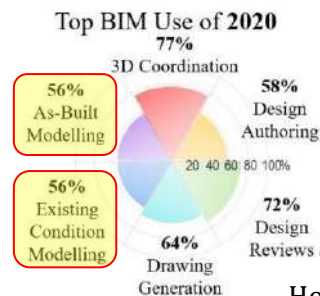
- ◆ 3D LiDAR scans in top BIM uses

- Surveys **actual** 3D dimensions, **real** assets
- **Every** developer/contractor has a laser scan team
- Affordable data sources for micro-scale urban models

BIM Use	Investigation, Feasibility and Planning	Design
1 Design Authoring	M ^B	M
2 Design Reviews	M ^B	M
3 Existing Conditions Modelling	M ^B	M
4 Site Analysis	M ^B	M
5 3D Coordination		M
6 Cost Estimation	O	M ^B
7 Engineering Analysis		M ^B
8 Facility Energy Analysis	O	
9 Sustainability Evaluation	O	M ^B
10 Space Programming	O	M ^B
11 Phase Planning (4D Modelling)		M ^B
12 Digital Fabrication		M ^B
13 Site Utilization Planning		
14 3D Control and Planning		
15 As-Built Modelling		
16 Project Systems Analysis		
17 Maintenance Scheduling		
18 Space Management and Tracking		
19 Asset Management		M ^B
20 Drawing Generation (Drawing Production)		M



BIM mandatory stages
(DevB TC(W) No. 2/2021)



Scan-to-BIM
(Moon Palace,
Src: authors)

Top BIM uses in
Hong Kong (Src: CIC)

(*: Est. figure by GovHK)



1 Challenges ahead



Three major challenges

- Hard-to-process LiDAR data

Large-scale, occluded, cluttered/noisy, unstructured

Worsened by >40 semantic labels

DL's point-object accuracy gap

>80% mIoU for points

<40% mIoU BIM objects

- 'Landing' for local industry

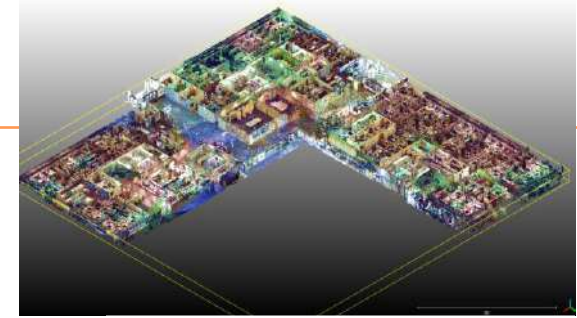
Construction business

Government's regulations

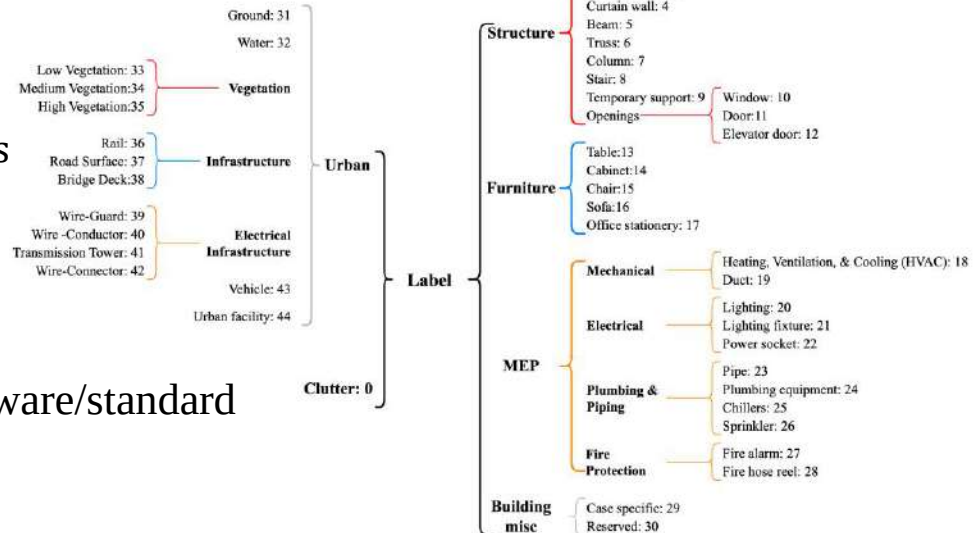
- BIM dominance by commercial software/standard

By A**/B**/C** companies

Gov./SME/Univ. may be reluctant



200 Mil points, 4,000 m² floor area

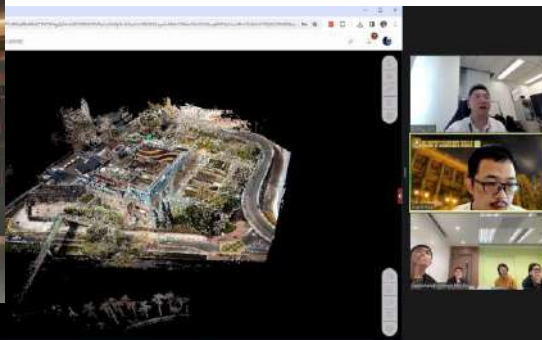




1 Visiting sites in HK, Shenzhen, Huizhou, Dongguan, Shanghai, etc.



ADCH





1 Needs for scan-to-BIM in practice



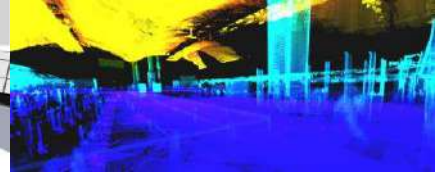
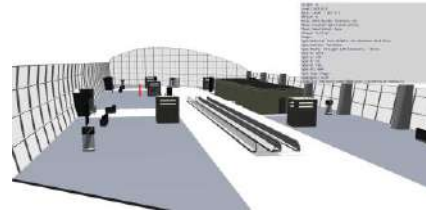
◆ Scan-to-BIM aims to

- **Verify** as-designed BIM
E.g., Real assets of HKIA terminal
- **Digital twin** built assets
E.g., Actual dimension/HVAC (retrofitted by tenants)
- **City model** of site/indoor/outdoor conditions

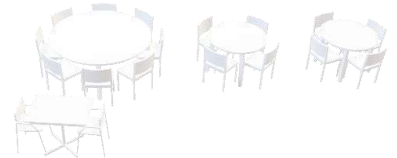
◆ Pain points felt by practioners

- **Inaccurate** manual segmentation
- **Manual** reconstruction of general AEs
- **No** 3D BIM objects
- **Low** applicability

By BIM consultant/regulator/planner



(Src: authors, Geosys)



(Src: authors) 8



1 General workflow; existing paradigms' scopes



◇ 4 steps

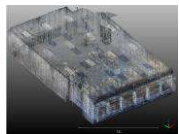
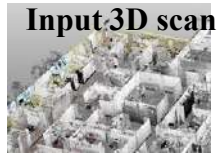
1.1 Point-level

1.2 Primitive-level

1.3 3D BIM details

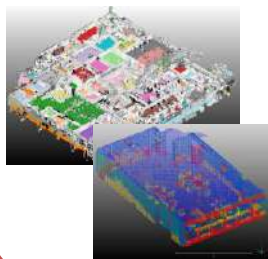
2. Applications

Input 3D scan



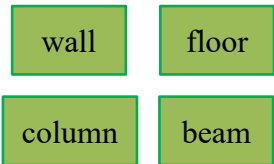
(Src: Authors)

1.1 Segmentation



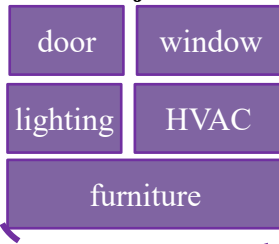
Time cost: ~5%

1.2 General architectural elements (AEs)



Time cost: ~20%

1.3 Detailed BIM objects



Time cost: ~75%

2. Applications

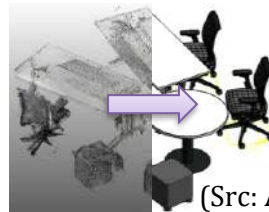
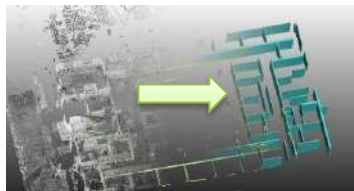
BIM/
Open-
BIM



CIM



(Src: Authors)



(Src: Authors)

1. mesher + *ifcopenshell*

Not covered

Not covered

2. Current AI software

Not covered



2 The SBASE system

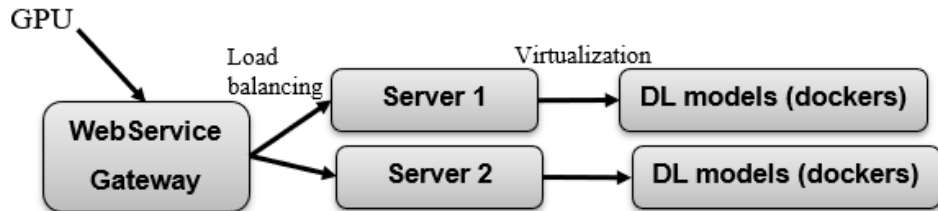


SBASE system



- With our award-winning algorithms

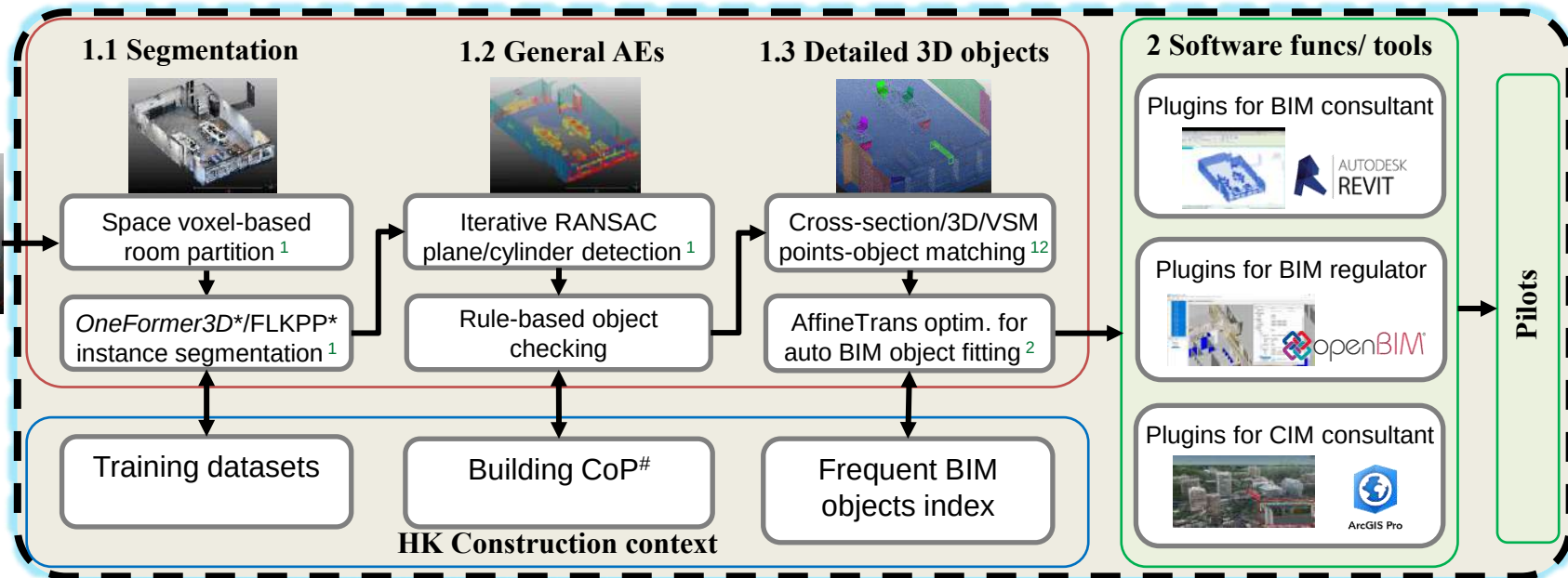
Process; 1.5GB points → 100M sample (1-cm voxel) → Web server



Server-side process (about 15 mins) : 'work' docker → convert to .npy
→ called trained DL models → convert back → Downloadable



Input:
3D points



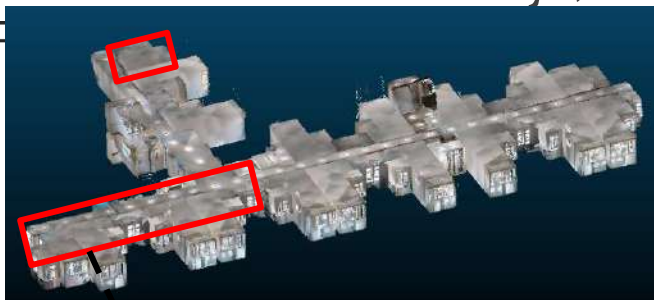
*: Or other commercialization-friendly DL; #: Code of Practice (BD, EMSD, WSD)



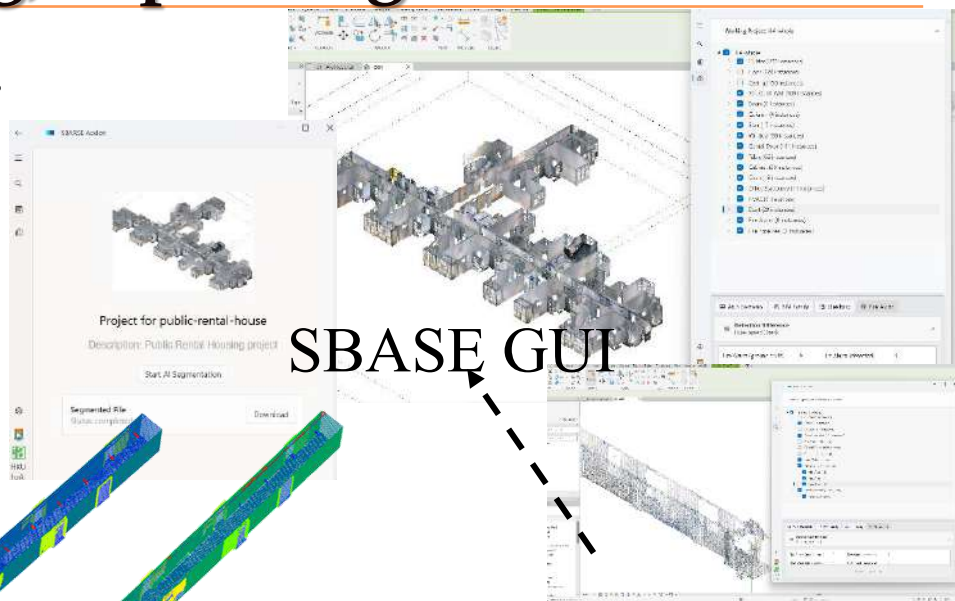
2 Case of public housing, Yip Wong Road Phase 1



◆ Data source: Geosys, and HKHA.



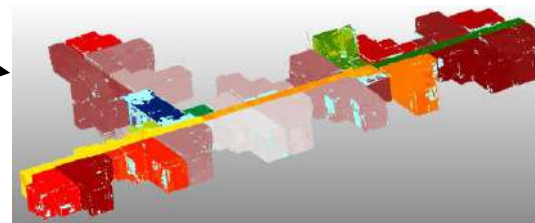
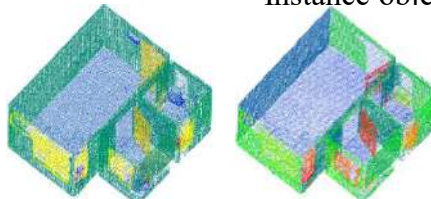
41,000,000 points in 2.5GB file



SBASE GUI

Point-level Label

Instance objects



Semantics + Instances



Scan-to-BIM metrics from pilots



◆ A reference floor scan of 12 offices.

NDCHK

KPI	Native Revit® (Ver 22.1)	Other paradigms (Aurivus® Ver 2)	Milestone goal^ (stress-test ver.)	Current
Overall performance (a floor of 30 case offices#)	200 labor-hours	140 labor-hours (30% saved)	100 labor-hours (50% saved)	✓
Input file size	--	20 GB	40+ GB	✓ Server ram
F_1 accuracy for General AEs	--	60~80%	70%	✓ 74%
F_1 accuracy for BIM objects	--	30% ^{^*}	40%	✓✓ 55%
3D BIM objects detection	--	N.A.	30%	✓✓ 35~80%

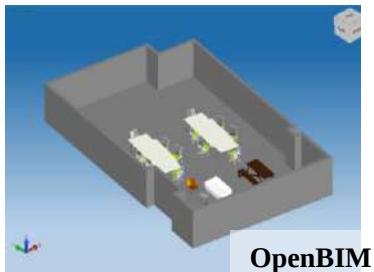


2.2. Light-weight CIM export



◆ Source: Student Residence at Wong Chuk Hang Site.

- ◆ 1. Remove costly objects by triangles/m³
- ◆ 2. shape-preserving QEM simplification



122,453 facets (.ifc)

Near-lossless
simplification



LoD4: 5,780 or 5% facets (of .ifc)
(with furniture)



Whole building output: [LoD-3 GIS model] <1.5% of BIM facets (no furniture)



3. Findings



◆ The 3 challenges partly solved by SBARSE



- LiDAR processing

 - Recent DL models (testing UniDet3D and some 2025 models)

 - Human modeler's functions to correct DL's FN/FP errors.

- 'Landing' for local industry

 - Objects enabled rule-based checking

 - Default checkings for local Code of Practice (CoP)

- Dominance by commercial

 - OpenBIM (BlenderBIM) / OpenGIS (Cesium) supports



3 Awards & news



ADCH





4. Disseminations to academia and industry



Secretary for Housing (Winnie) at CIC Expo 2023, Sky100



HKU FEng. Alumni
Assoc., iLab, 2023



2024: LSCM Summit, Hong Kong



LSCM Keynote, 2023 China Smart Construction Conference



Conferences in Greece,
Tsinghua, & SCUT





4. Extensions and future work

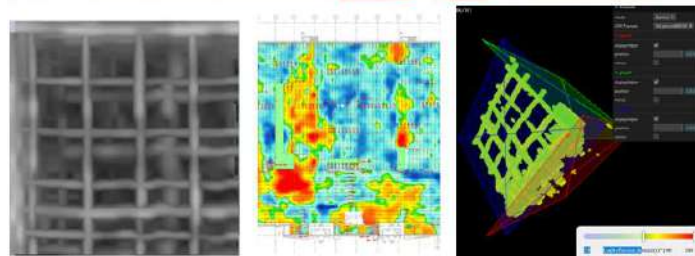
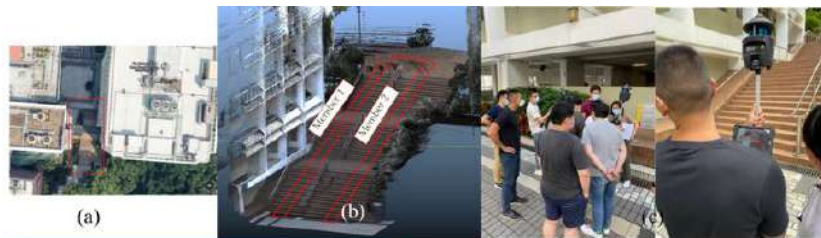


◆ Deep Feature/GenAI 3D modeling (patent)



◆ Teaching

◆ Integration with BIM, concrete GPR scan





5 Summary



- ◆ Smart construction, as a part of smart city, is needed in HK and the world
- ◆ Scan-to-BIM is vital to smart construction
- ◆ DL/AI is helpful in
 - Automating Scan-to-BIM
 - Compliance checking of objects
 - Compression of excessive model details
- ◆ Still much to improve
 - DL accuracy (instance-level is lower than expected)
 - GenAI may reveal new directions
 - Multi-source can enrich the scan-to-BIM
- ◆ Looking forward to joint efforts with multi-disciplinary research



5 Acknowledgements



◆ Funding: ITC ITF (ITP/004/23LP)

◆ Lead applicant: LSCM Center

◆ Research team

Name	Position	Remark	Role
Anthony Yeh	Co-PI		CIM functions and pilot
Weisheng Lu	Co-PI		BIM pilot
Ke Chen	Co-I	Mainland	liaison and promotion (National Center for Smart Construction)
Yijie Wu	Co-I		Point cloud algorithm; optimization
Maosu Li	Co-I		ArcGIS Pro
Zhe Chen	Co-I		Documentation, BPA
Dong Liang	Co-I		Point cloud semantic segmentation
Jiajia Wang	Co-I		3D features extraction for existing buildings

◆ Sponsors

- GeoSys , Unotech

◆ Collaborators

- XenseTech,, CIC, HKABAEIMA
- HKICBIM, HKIBIM, HKGISA,
- CSC-Hailong, HUST, OkayGIS

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Hong Kong Alliance of Built Asset & Environment
Information Management Associations
香港建設資產及環境信息管理聯盟



中運海龍科技有限公司
CHINA STATE CONSTRUCTION HAILONG TECHNOLOGY COMPANY LIMITED



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*AI saves BIMers and creates
new values*

Thank you!



References



- ◇ BD. (2020). Electronic Submission Hub for centralised processing of electronic submission of building development and other applications under the Buildings Ordinance. https://www.bim.cic.hk/en/bim_showcases/bim_talks_detail/48?back=%2Fzh-hans%2Fbim_showcases%2Fbim_talks
- ◇ CIC. (2022). CIC-BIM Network Sharing Session. Hong Kong: CIC-BIM
- ◇ DevB. (2019). Adoption of Building Information Modelling for Capital Works Projects in Hong Kong. DEVB TC(W) No. 9/2019 <https://www.devb.gov.hk/filemanager/technicalcirculars/en/upload/371/1/C-2019-09-01.pdf>
- ◇ DevB. (2021). Adoption of Building Information Modelling for Capital Works Projects in Hong Kong. DEVB TC(W) No. 2/2021 <https://www.devb.gov.hk/filemanager/technicalcirculars/en/upload/387/1/C-2021-02-01.pdf>
- ◇ Li, M., Xue, F., Wu, Y., & Yeh, A. G. (2022). A room with a view: Automatic assessment of window views for high-rise high-density areas using City Information Models and deep transfer learning. *Landscape and Urban Planning*, 226, 104505.
- ◇ MV. (2022). *Private communication*. Hong Kong: Market vendor.
- ◇ Thomas, H., Qi, C. R., Deschaud, J. E., Marcotegui, B., Goulette, F., & Guibas, L. J. (2019). Kpconv: Flexible and deformable convolution for point clouds. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 6411-6420).
- ◇ Xue, F., Lu, W., Chen, K., & Zetkovic, A. (2019). From semantic segmentation to semantic registration: Derivative-Free Optimization-based approach for automatic generation of semantically rich as-built Building Information Models from 3D point clouds. *Journal of Computing in Civil Engineering*, 33(4), 04019024.
- ◇ Xue, F., & Lu, W. (2020). A semantic differential transaction approach to minimizing information redundancy for BIM and blockchain integration. *Automation in construction*, 118, 103270.
- ◇ Xue, F., Lu, W., Chen, Z., & Webster, C. J. (2020). From LiDAR point cloud towards digital twin city: Clustering city objects based on Gestalt principles. *ISPRS Journal of Photogrammetry and Remote Sensing*, 167, 418-431.
- ◇ Xue, F., Wu, L., & Lu, W. (2021). Semantic enrichment of building and city information models: A ten-year review. *Advanced Engineering Informatics*, 47, 101245.
- ◇ Xue, F. (2022). As-built building information modeling: joint effort of 3D reconstruction and semantic enrichment. In *Research Companion to Building Information Modeling*. Edward Elgar Publishing.